

Journal Inquiry

Collaborative Development Design

ID 68620

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Journal Inquiry

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Journal Inquiry

Introduction

There are occasions when a change to a database table needs to be investigated. With all of the tables in the HarrisData database being journaled, provide the ability to extract the information stored in the journal receivers to allow the user to research the changes to a table.

Objective

Add the following options to the HTML interface:

- Load Journal Inquiry
 - Allows the user to select the database table and timeframe to extract data from the journal receiver. This option can be processed immediately or scheduled to run as needed.
 - Creates a Journal Inquiry (J_) database table that can be queried by the user.
- Journal Inquiry
 - Allows the user to display or download the information extracted from the Journal Inquiry (J_) table to a csv.

Scope

This project will be designed and delivered with the following release and version levels:

HarrisData HD5.0 115166
IBM i V7R1M0

This project will be based on the HTML User Interface. The Legacy User Interface will continue to work as it does today.

Minor changes may be made to the design as the enhancement is coded. These changes are made to correct issues discovered in developing the enhancement that were not found during the design. If any changes are made they will be included in the copy of the design you receive with the final product delivery. This design will also serve as your documentation until the User Documentation is updated.

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Programming Requirements

Load Journal Inquiry

The Load Journal Inquiry is accessed from the Utilities->Control Panel Plus page. It requires 'Load' authority in Program Option Security for program 'JOURNALINQ'.

The Load Journal Inquiry (Figure 1) provides the following functionality:

- Required values:
 - Table Name – must exist in the library list and be journaled
 - Number of Days – must be greater than zero
- Allow the job to be scheduled (Figure 2)
- Icons:

	Returns user to the Utilities->Control Panel Plus page
	Submit/Schedule the job
	Displays the Schedule Job options
	Clears all entries
	Displays the Journal Inquiry design in a separate tab

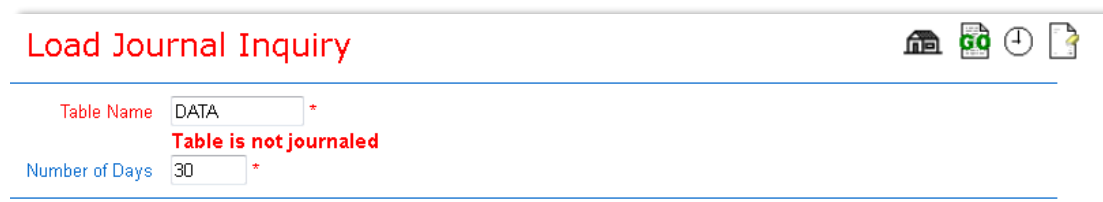
- Use the Display Journal (DSPJRN) command for the Table Name entered to retrieve entries from the journal receivers based on the following criteria:
 - Journal Code 'R' - which contains information about a change to a specific record in the physical file
 - Journal Entry Types included are:
 - 'DL' Delete
 - 'PT' Add
 - 'UB' Update Before
 - 'UP' Update After
 - Start Date = current date minus Number of Days
- The entries extracted from the DSPJRN command will be updated to a Journal Inquiry (J_) table with the following naming convention:
 - If the Table Name is less than 8 characters, 'J_' will be appended at the beginning:
 - HDIWHS -> J_HDIWHS
 - If the Table Name is greater than 8 characters, 'J_' will replace the first 2 characters:
 - HDIMSTEXT -> J_IMSTEXT

Note: The Journal Inquiry (J_) table data will be replaced each time the Load Journal Inquiry option is run.

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- The Journal Inquiry table (J_) will be created in the same database where the Table Name was found. It will contain the following columns:
 - Columns from the DSPJRN command include:
 - Sequence
 - Type
 - Timestamp
 - Job Name
 - User
 - Program
 - All columns from the Table Name entered will be included.
 - Refer to the Database Requirements section for an example.
- After the Journal Inquiry (J_) table is created, it will be used to create a csv file in the attachments under folder name 'JournalInquiry'. This is an example of the name of the csv file which contains the following information:
 - **ItemWarehouse_2019-10-21_to_2020-06-17_(J_HDIWHS)**.csv
 - **Table Name Description**
 - **Start Date to Current Date**
 - **Journal Inquiry (J_) table name**

Note: If the csv file already exists, it will be deleted and re-created. This would only occur if the same Table Name and Number of Days was loaded more than once on the same date resulting in the same csv file name.



Load Journal Inquiry

Table Name *

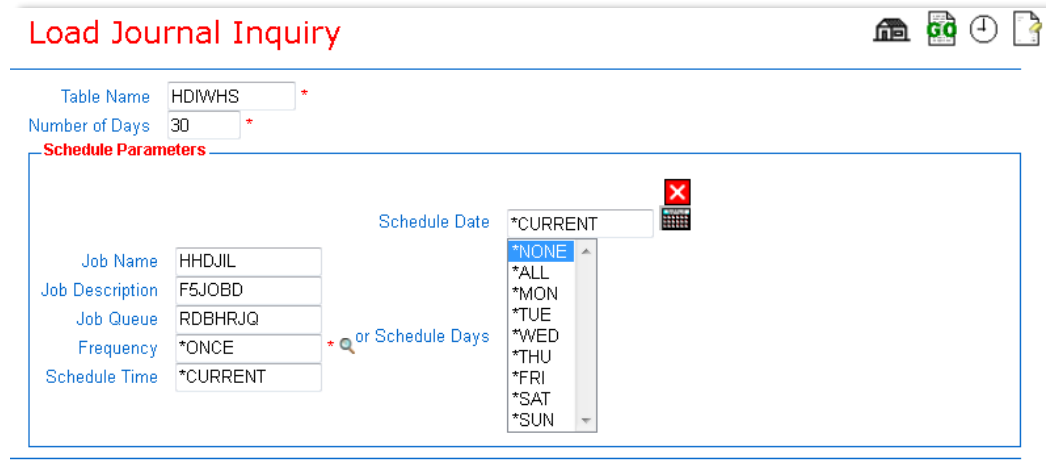
Number of Days *

Table is not journaled

Figure 1

Journal Inquiry

The Load Journal Inquiry page (Figure 2) displays the standard Schedule Job parameters.



The screenshot shows a web application window titled "Load Journal Inquiry". At the top right, there are icons for home, go, clock, and a document. Below the title bar, there are two input fields: "Table Name" with the value "HDIWHS" and "Number of Days" with the value "30". Both fields have a red asterisk to their right. Below these fields is a section titled "Schedule Parameters" in red. Inside this section, there are several input fields and a dropdown menu. On the left, there are five fields: "Job Name" (HHDJIL), "Job Description" (F5JOB0), "Job Queue" (RDBHRJQ), "Frequency" (*ONCE), and "Schedule Time" (*CURRENT). To the right of these fields is a red asterisk and a magnifying glass icon, followed by the text "or Schedule Days". Further right is a "Schedule Date" field with the value "*CURRENT" and a calendar icon. To the right of the "Schedule Date" field is a dropdown menu with a red 'X' icon above it. The dropdown menu is open, showing a list of options: *NONE, *ALL, *MON, *TUE, *WED, *THU, *FRI, *SAT, and *SUN. The *NONE option is currently selected.

Table Name	HDIWHS
Number of Days	30
Schedule Parameters	
Job Name	HHDJIL
Job Description	F5JOB0
Job Queue	RDBHRJQ
Frequency	*ONCE
Schedule Time	*CURRENT
Schedule Date	*CURRENT
or Schedule Days	*NONE, *ALL, *MON, *TUE, *WED, *THU, *FRI, *SAT, *SUN

Figure 2

Journal Inquiry

Journal Inquiry

The Journal Inquiry page (Figure 3) is displayed when the Journal Inquiry option is selected from the Export portal and provides the following functionality:

- Icons:

	Download the csv file locally
	Deletes the csv file. This icon is only displayed if the user has 'Delete' Program Option Security for program 'JOURNALINQ'. A confirmation box (Figure 4) will display prior to the delete.

- Displays a list of existing Journal Inquiry csv files created by the Load Journal Inquiry in descending Created order. The csv files are stored in the attachments in folder 'JournalInquiry' with a path similar to:
 - /HarrisData/EIP/Attachments/DB/JournalInquiry/
- The File Name is a link to display the raw csv data in an html table format (Figure 5).
- If no journal entries are found for the Table Name and Number of Days selected, the File Name will start with 'NO JOURNAL ENTRIES FOUND' and the csv file cannot be displayed or downloaded.

Journal Inquiry

Opt	File Name	Created
	 ItemWarehouse 2020-02-20 to 2020-06-19 (J_HDIWHS).csv	June 19 2020 07:45:00.
	NO JOURNAL ENTRIES FOUND -Item 2020-05-20 to 2020-06-19.csv	June 19 2020 07:44:00.
	 ItemWarehouse 2019-10-21 to 2020-06-17 (J_HDIWHS).csv	June 17 2020 13:36:13.
	 ItemPlant 2020-02-14 to 2020-06-13 (J_HDIPLT).csv	June 17 2020 13:15:48.
	 ItemWarehouse History 2019-10-21 to 2020-06-17 (J_HDISLH).csv	June 17 2020 13:15:38.

Figure 3

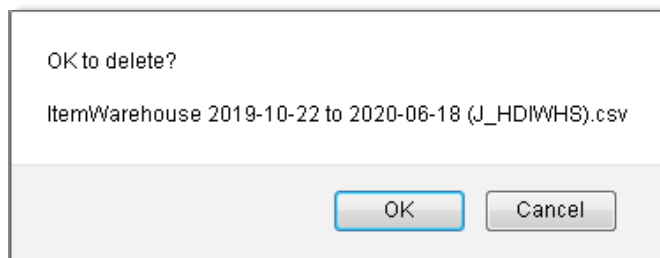


Figure 4

Journal Inquiry

The html display of the csv file (Figure 5) will include:

- File Name
- A legend explaining the different types of records included:
 - (Type: DL=Delete, PT=Add, UB=Update Before, UP=Update After)
- All columns from the csv file.
- No formatting of data. It will be displayed as it is stored. So data such as dates could display as CYMD (1200603) or ISO 2020-06-03.
- No sorting/filtering/paging options.
- Use the standard browser Find (ctrl F) option to search for data.

Journal Inquiry														
ItemWarehouse 2019-10-22 to 2020-06-18 (J_HDIWHS).csv														
(Type: DL=Delete, PT=Add, UB=Update Before, UP=Update After)														
Sequence	Type	Timestamp	Job	User	Program	Inact Code	Item Number	WH	Price UOM	Purch UOM	Piece Per Price	Piece Per Purch	Piece Per Case Pack	Stock Locator Active
1746785	PT	2020-01-22-12:29:47.021312	QPADEV0003	RDBHRPR	HHDIWU_E		VSP-323-3	11	EA	EA	1.0000	1.0000	.0000	N
1772093	UB	2020-01-24-06:55:53.240912	F5_STROEDQ	RDBHRPR	HOEUOE		VSP-323-3	11	EA	EA	1.0000	1.0000	.0000	N
1772094	UP	2020-01-24-06:55:53.240912	F5_STROEDQ	RDBHRPR	HOEUOE		VSP-323-3	11	EA	EA	1.0000	1.0000	.0000	N
1772170	PT	2020-01-24-06:58:05.778576	QPADEV0003	RDBHRPR	HHDIWU_E		VSP-323-5	0	EA	EA	1.0000	1.0000	.0000	N
1772172	PT	2020-01-24-06:58:05.784256	QPADEV0003	RDBHRPR	HHDIWU_E		VSP-323-5	11	EA	EA	1.0000	1.0000	.0000	N
1772183	DL	2020-01-24-06:58:25.579904	QPADEV0003	RDBHRPR	HHDIWU_E		VSP-323-5	0	EA	EA	1.0000	1.0000	.0000	N

Figure 5

Journal Inquiry

Program Option Security

A program name of 'JOURNALINQ' will be added to Security->Program Option Security to control processing of the Journal Inquiry with the following options:

- Load – user can process the *Load Journal Inquiry* option on the Utilities->Control Panel Plus page.
- Delete – user can delete csv files from the *Journal Inquiry* page accessed from the Export portal.

Once the users are setup that requires the above authorities, you should add the 'HDS' user profile with no options selected to restrict all other users.

Program Option Security Maintenance

Program Name: [JOURNALINQ]

User Profile	Name	Role

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Opt	User Profile	Name	Role	Load	Delete
	HDS	HDS	HD_ALL	☐	☐
	RDBHRPR	Brian	RD_ALL	☒	☒
	SPMCMPR	Matt	RD_ALL	☒	☒

Journal Inquiry

Database Requirements

The following is an example of a Journal Inquiry (J_) table that is created by the Load Journal Inquiry option. It contains **columns from the DSPJRN command** followed by all of the columns for the selected Table Name.

Example: Item/Warehouse History (J_HDISLH)

Column	Description	Start	End	Bytes	Dc	T
JOSEQN	Sequence	1	10	10	0	S
JOENTT	Type	11	12	2		A
JOTSTP	Timestamp	13	38	26		Z
JOJOB	Job	39	48	10		A
JOUSER	User	49	58	10		A
JOPGM	Program	59	68	10		A
IHITEM	Item Number	69	83	15		A
IHWHS	Warehouse Number	84	86	3	0	S
IHPERD	Transaction Period	87	91	5	0	S
IHCYR	Transaction Year	92	94	3	0	S
IHQREC	Quantity - Received	95	101	13	4	P
IHQAMT	Quantity - Adjustments	102	108	13	4	P
IHQTRF	Quantity - Transferred	109	115	13	4	P
IHQTYI	Quantity - Issued	116	122	13	4	P
IHQKTI	Quantity - Kit Issued	123	129	13	4	P
IHQRTV	Quantity - Returned To Vendor	130	136	13	4	P
IHQSCR	Quantity - Scrapped	137	143	13	4	P
IHQSLD	Quantity - Sold	144	150	13	4	P
IHQDSH	Quantity - Drop Shipped	151	157	13	4	P
IHDLRS	Dollars - Sold	158	164	13	2	P
IHSTOT	Number Of Stockouts	165	169	9	0	P